Gear Comp

User manual and parameter reference · 0.1.1

An analog-style compressor with program-dependent release, sidechain filtering and parallel mix.

Quick start

1. Insert Gear Comp on a bus or track and load Gentle Glue.
2. Set Threshold and Ratio while watching GR; adjust Attack and Release to the material.
3. Use Mix for parallel compression and either set Output or arm Auto Gain and edit a sound knob.

Editor window

960 × 300 minimum; 1200 × 388 initially, resizable. Dimensions are logical editor pixels, before host window chrome. HELP adds 92 pixels of height (Gear adds a one-pixel divider too). Allow enough usable screen space at your display scale. Press HELP again to close the help area.

Operation

Output is wet-path makeup before Mix. The meter switch only changes the display and never starts learning. The sidechain filter changes the detector, not the audio frequency response.

The resizable editor keeps all header actions visible at its supported minimum size. HELP explains the focused or hovered part.

Auto Gain is an armed workflow, not continuous normalization. Only direct user changes to sound knobs start learning while it is enabled; it holds for eight seconds after the last such edit and then idles. Output is excluded so you can trim it while idle. Turning Auto Gain on only arms it. Switching it off stops learning. Presets do not store Auto Gain; session recall retains the switch and Output but starts idle. Offline rendering uses the saved output.

Controls and operation

THRESHOLD

The level above which the compressor starts to turn the signal down, -40 to 0 dB. The lower it is, the more of the signal is over it and the more is compressed. How it eases in around it is the Knee switch's.

RATIO

How much of what goes over the threshold is let through: at 4:1, 4 dB over the threshold comes out 1 dB over it. 1:1 does nothing; 20:1 is close to a limiter. Low ratios are gentle and transparent, high ones squash.

ATTACK

How fast the compressor reacts once the signal goes over the threshold, 0.1 to 100 ms. Slow lets the first bite of a drum or a pluck through; fast catches it and makes the sound flatter and closer. The level it follows is an analog-style detector: a peak follower with an RMS floor, so a fast attack does not make the bass distort.

RELEASE

How long it takes to let go once the signal falls back under the threshold, 50 ms to 2 s. It follows the program: after a short burst it lets go about three times faster than set, after sustained compression it takes the full time, as an opto cell or a tube compressor does. Short releases follow the music and pump; long ones are smooth.

OUTPUT

Makeup gain after the compressor, 0 to +20 dB, to bring back the level that compression took off. Compare with Power off to judge the sound, not the loudness, or let Auto Gain set it.

MIX

The compressed signal against the dry one, 0 to 100%, after the makeup: parallel compression. Heavy compression mixed back under the dry signal keeps the punch and gets the body; the dry signal is never touched by Threshold or Output.

SIDECHAIN

A high-pass in the detector only, never in the sound: Off, 60 Hz or 120 Hz (12 dB an octave), as the switchable filters of classic bus and mastering compressors. Bass-heavy material pushes a compressor hard; with the filter on the bass is still heard but no longer drives the gain reduction, so the mix stops pumping on every kick. Off by default, as the hardware's is.

KNEE

How the compressor eases in around the threshold: Hard (a corner, as a fast peak compressor bites), Soft (6 dB, the default), Wide (12 dB, a gentle, transparent start).

AUTO GAIN

Arm Auto Gain, then directly edit a sound knob to match loudness by turning Output. It learns for eight seconds after the last user knob edit, then holds the result while still on. Switches, choices, host automation, presets and session recall do not start learning. Output itself is a manual trim while idle; it is protected during learning. Auto Gain is excluded from presets. Session recall starts idle, and offline rendering uses the saved Output. Working has a pulsing dot and time-left line; idle has a steady dot.

VU METER

Gain reduction or the output level, as the GR / OUT switch says. As gain reduction it reads how many dB the compressor is taking off, from 0 VU downward; as output, 0 VU is -18 dBFS RMS. The needle moves as a VU meter does (99% of a step in 300 ms).

GR / OUT

What the meter shows: GR, the gain reduction (how hard the compressor is working), or OUT, the output level.

GR 1 dB

Lit while the compressor takes off 1 dB or more.

GR 2 dB

Lit while the compressor takes off 2 dB or more.

GR 4 dB

Lit while the compressor takes off 4 dB or more: it is working.

GR 8 dB

Lit while the compressor takes off 8 dB or more: it is working hard.

POWER

Switches the compressor on or off. Off plays the input as it is.

POWER LAMP

Lit while the compressor is on.

PRESETS

Browse, load and save settings. An asterisk means the sound has changed since the preset was loaded.

HELP

Shows or hides this bar. It says what the control under the mouse is and what it does.

BYPASS

The same as the POWER switch: plays the input as it is.

Every parameter

Values and defaults are taken from the plugin. A control's effect can depend on the selected mode; see the operation notes above. Choices keep their saved identity between versions. Knobs and switches can be automated unless marked otherwise.

Parameter	What it does	Values and default
Threshold	Threshold: -18.0 dB	-40.0 dB to 0.0 dB Default: -18.0 dB
Ratio	Ratio: 4.0:1	1.0:1 to 20:1 Default: 4.0:1
Attack	Attack: 10 ms	0.10 ms to 100 ms Default: 10 ms
Release	Release: 200 ms	50 ms to 2000 ms Default: 200 ms
Output	Output: 0.0 dB	0.0 dB to +20.0 dB Default: 0.0 dB
Meter	Meter: Gain Reduction	Gain Reduction, Output Default: Gain Reduction Not host automatable
Mix	Mix: 100%	0% to 100% Default: 100%
Sidechain	Sidechain: Off	Off, 60 Hz, 120 Hz Default: Off Not host automatable
Knee	Knee: Soft	Hard, Soft, Wide Default: Soft Not host automatable
Auto Gain	Keeps the compressed signal as loud as the dry input by turning the Output knob: arms when turned on; only direct sound-knob edits start eight seconds of learning, then it idles	Off to On Default: Off
Bypass	Plays the input as it is (the POWER switch on the unit)	Off to On Default: Off

Presets and files

Presets are browsed with the arrows or the preset-name menu. A star marks an edited patch. Save supports folders, Rename, Delete and opening the user preset folder; dropping a compatible preset file onto the bar loads it. A newer-file indication means this version could not apply every setting.

The host session stores the complete patch. A product preset can deliberately leave listening trim, audition or display controls unchanged; the product-specific rules below describe those exceptions.

Gear Comp presets use .gearcomp in ~/Library/Audio/Presets/Barakigal Audio/Gear Comp. Power/Bypass and Auto Gain are excluded from presets; Output belongs to the processing sound and is included.

Installation and help

Download the macOS archive from the product page and quit your audio host. Place the AU .component in ~/Library/Audio/Plug-Ins/Components, VST3 in ~/Library/Audio/Plug-Ins/VST3, or CLAP in ~/Library/Audio/Plug-Ins/CLAP. The product page also offers an installer with format selection. Restart or rescan the host. To uninstall, quit the host and remove that product's bundles from those folders; saved sessions and user presets are separate.

For questions or a problem report, use BRK Audio support on GitHub. Include the plugin version, macOS version, host and format, sample rate and a short description of how to reproduce it.